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STRAIN, MAGNETIC FABRIC, AND PALEOMAGNETISM OF THE DEFORMED RED BEDS OF THE PONT-REAN FORMATION, BRITTANY, FRANCE

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Abstract. A joint study of strain, anisotropy of magnetic susceptibility (AMS), and paleomagnetism has been conducted in the deformed red beds of the Ordovician Pont-Réan formation, Brittany, France. Analysis of results from 13 sampling sites shows that homogeneous strain, estimated from the shape of elliptical reduction spots in each site, controls the development of a magnetic fabric during the Hercynian folding of this formation. The good correlation between strain and AMS tensors is interpreted as mainly arising from a strain-induced progressive rotation of planar hematite that tends to align within cleavage plane. This rotation of magnetic carriers in turn induces a rotation of net measurable pretectonic remanent magnetization toward cleavage. Because of the high angle between the initially subvertical Ordovician remanent magnetization and the regionally subhorizontal shortening direction, this deviation of the pretectonic magnetization is achieved without any scattering but with a probable clustering effect upon within-site initial dispersions. For the same reason the overall distribution of mean-site pretectonic magnetization vectors is clustered at the in situ deformed state. This results in an anomalous behavior of the classical fold test, that could erroneously lead to assignment of a syntectonic to post tectonic age to the magnetizations. Finally the strain removal technique based upon the passive behavior hypothesis is applied to the deviated vectors. It allows recovery of the initial direction of these pretectonic Ordovician magnetizations ($D = 235^{\circ}$, $I = 75^{\circ}$, $\alpha_{95} = 6.5^{\circ}$).

Introduction

A basic idea, commonly admitted by all researchers in the field of paleomagnetism, is that if a magnetized sedimentary bed is folded, its magnetization will be deflected away from its original direction. Consequently, if one wants to determine the original direction of magnetization within a folded area, he must first recover the rotation or sequence of rotations that have affected each paleomagnetic vector studied. The more widely used practice to this purpose is to consider that the geometry of the folded bedding surface, assumed to be initially planar and horizontal, provides good indicators of these rotations. From the description of the possible rotation(s) undergone by the bedding surface at each site of paleomagnetic sampling, an overall mean direction of pretectonic magnetization can then be recovered by applying the inverse rotation(s) that should unfold the rock formation. However, this treatment, known as the "(bedding-) tilt correction" [Graham, 1949], basically relies on the implicit assumption that the rotations of bedding plane actually describe the rotations of paleomagnetic vectors. Clearly, this is true only if the magnetic vector remains unchanged within the bedding reference frame throughout the folding of rock unit, that is, when folding is achieved by rigid buckling of the beds, without any internal deformation. In effect, among other grain-scale deformation mechanisms, it is now established that a strain-induced rotation of magnetic carriers (e.g., see X ray diffractometry measurements by Cogné and Gapais [1986]) can produce significant deviations of magnetic vectors. This has been shown by analogical simulations [Ozima, 1980; Cogné, 1987b; Anson and Kodama, 1987], numerical models [Cogné et al., 1986; Van der Pluijm, 1987; Kodama, 1988] and through paleomagnetic analysis of deformed red beds [Kligfield et al., 1981, 1983; Cogné and Perroud, 1985; Cogné, 1987a]. This strain-induced deviation of paleomagnetic vectors may be considered as a second kind of rotation, with respect to the internal reference frame of bedding plane. Consequently, in the case of folding with internal deformation (for example, when cleavage develops) the description of paleomagnetic vector rotations in terms of bedding surface geometry is incomplete, and thus conclusions drawn from the behavior of paleomagnetic vector populations with respect to the classical fold test may be inadequate. This has been clearly demonstrated on the basis of some numerical models by Van der Pluijm [1987] and Kodama [1988], and by Cogné and Perroud [1985] in the deformed red beds of the Maritime Alps (France).

The present paper addresses two main points: (1) to document the characterization of strain effects upon pretectonic remanent magnetization through the study of strain, anisotropy of magnetic susceptibility (AMS), and paleomagnetism within deformed red beds, and (2) to defend the idea that a correction for strain-induced deviations of paleomagnetic vectors is possible within such formations [Cogné and Perroud, 1987]. This work is the third example in a series of similar investigations we have previously published, first in the Permian red beds of the Maritime Alps [Cogné and Perroud, 1985], then in the Permian red beds of the "Col du Somport" in the Pyrenean Chain [Cogné, 1987a].

Geological Setting and Sampling Sites

The studied area is situated in the central part of the Armorican Massif, Brittany, France (Figure 1) (mean position: 48°N, 2.5°W) and forms

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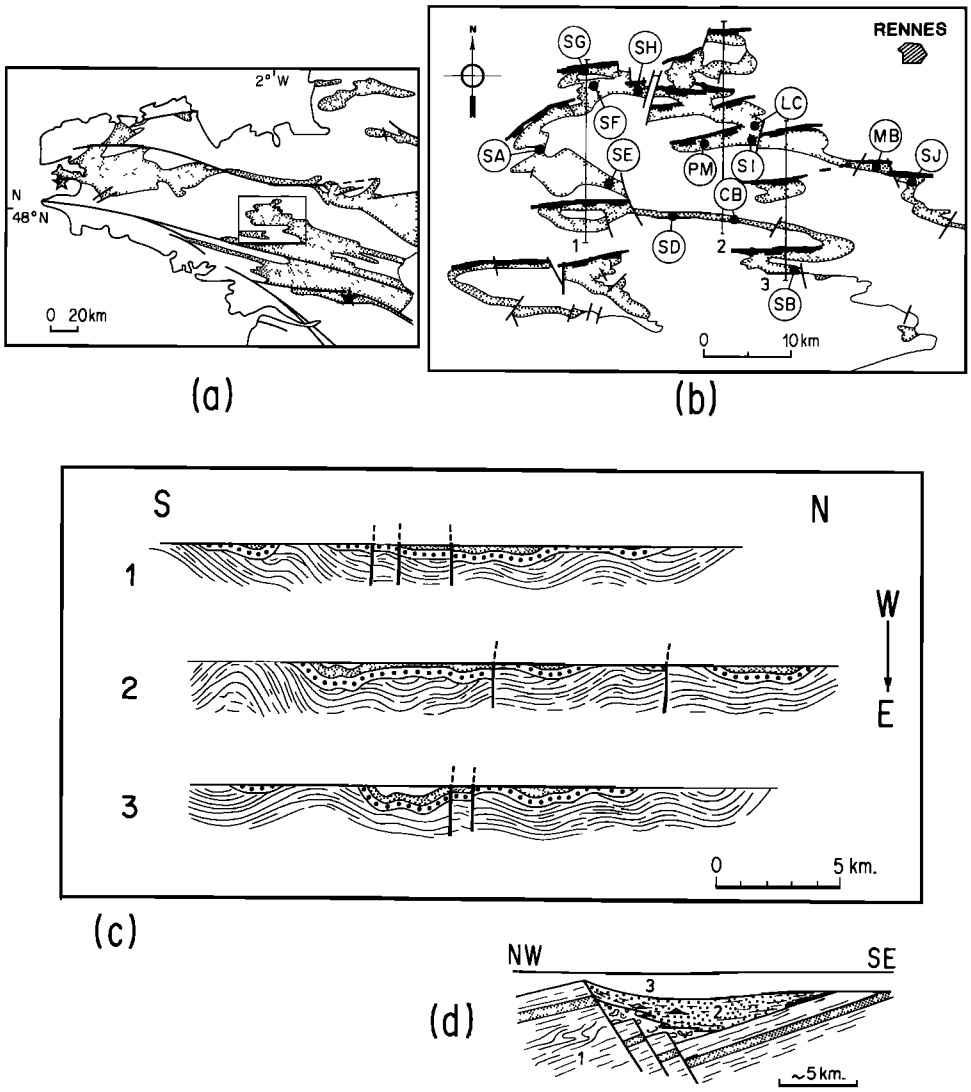


Fig. 1. (a) Location map of the studied area (boxed); dotted area, Paleozoic cover; open (solid) star: "Cap de la Chèvre" ("Moulin de Chateaupanne") Formation. (b) Location map of the sampling sites (labeled dots) around the syncline structure of the Pont-Réan Formation (dotted area). Heavy lines are synsedimentary normal faults. (c) Schematic N-S cross-sections 1 to 3 as located in Figure 1b; three formations are shown: the "Grès Armoricain" Formation (small dots), the Pont-Réan Formation (large dots), and the underlying folded Brioverian basement. (d) Schematic cross section illustrating the structural context during deposition of the Pont-Réan formation [after Ballard et al., 1986]; 1, Brioverian basement; 2, Pont-Réan Formation; 3, "Grès Armoricain" Formation.

part of multilayered sequence of Paleozoic sediments in the region (Figure 1a). The Pont-Réan red bed formation begins the Paleozoic sedimentation in Central Brittany, unconformably overlying the pelitic Brioverian basement, and overlain by the "Grès Armoricain" Formation. The Pont-Réan Formation has not yet been dated, but a paleontologically based Arenig (lower Ordovician) age has been proposed by Cavet et al. [1979] for the lateral equivalent "Moulin de Chateaupanne" formation, and a radiometric age of 465 +/- 1 m.y. by U-Pb method has been obtained by Bonjour et al. [1988] on volcanic intercalations within the lateral equivalent "Cap de la Chèvre" Formation in the Crozon peninsula (Figure 1a). It seems therefore reasonable to infer a lower Ordovician age for the Pont-Réan Formation. The major phase of the Hercynian orogeny in the Armoricain Massif is responsible for the folding of the Precambrian and Paleozoic formations of Central Brittany (Figures 1b and 1c) [Le Corre, 1978; Percevault and Cobbold, 1982]. For the Pont-Réan Formation, this results in an open syncline structure, with subhorizontal approximately E-W trending axes. Folding is accompanied by the development of subvertical E-W

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trending axial planar cleavage. Structural studies based upon strain marker measurements [Le Corre and Le Théoff, 1976], texture development [Gapais and Le Corre, 1981], and metamorphism [Hanmer et al., 1982] were conducted in Central Brittany. They showed a clear strain gradient from N to S in the region, leading Gapais and Le Corre [1980] and Percevault and Cobbold [1982] to attribute the Hercynian folding of the Paleozoic cover to a crustal dextral shearing process. Finally, recent investigations of Ballard et al. [1986] and J. P. Brun (personal communication 1987) based upon the study of Brioverian-Paleozoic unconformities, of thickness variations in the Pont-Réan and Grès Armoricain formations etc., led these authors to propose that deposition of the Pont-Réan Formation took place during an extensional event. The sediments were thus deposited upon Brioverian tilted blocks in small basins of half graben type (Figure 1d). One can note that near to normal faults, the bedding plane may have had a significant syn-deposition dip of 10° to 20° toward the center of the basin. We shall see later that this point is probably significant with respect to the final distribution of primary paleomagnetic directions. Note that a similar syn-sedimentary dipping was suspected by Diot [1980] in the Moulin de Chateaupanne Formation. This was confirmed by Perroud et al. [1986], who showed that the Ordovician paleomagnetic direction of this formation is more accurately determined before any tilt correction than after.

The paleomagnetic study of three hand samples from the Pont-Réan Formation allowed Duff [1979] to isolate a steeply dipping magnetization component. These very high inclinations are characteristic of the Ordovician magnetic field in the Armorican Massif (see compilation by Perroud [1985]). We could thus suspect the existence of a pretectonic remanent magnetization within the Pont-Réan red bed formation, which was verified by our preliminary results obtained in this formation [Cogné et al., 1986].

A total of 13 sites, including those of the preliminary study, have been sampled all around the syncline structure (Figure 1b). As noted in previous similar studies [Cogné and Perroud, 1985; Cogné, 1987a], in order to assign each paleomagnetic direction a given strain state, a "sampling site" has the stringent definition of the maximum rock volume in which strain can be assumed to be macroscopically continuous and homogeneous. Field criteria for strain homogeneity are (1) initially planar surfaces remain planar after deformation, (2) initially parallel planes remain parallel. There were 177 cores (25 mm in diameter, 50 to 100 mm long) drilled and oriented in situ using magnetic and sun compasses. Along with this paleomagnetic sampling, structural measurements (orientation and dip of bedding and cleavage planes, stretching lineation, etc.) were made at each site. As in other red formations, some small green elliptical zones, known as reduction spots, provided strain markers [e.g., Wood et al., 1976; Graham, 1978; Kligfield et al., 1981]. At least one oriented block containing such reduction spots (generally of millimetric dimensions) was sampled in each site (except SH) in order to estimate the mean strain values of the site.

Strain and Anisotropy of Magnetic Susceptibility

Techniques and measurements

Because of the small dimension of reduction spots, their axial ratio and orientation were measured on photographic enlargements of polished sections of blocks sawed following the three principal strain planes. These measurements have shown that within the limits of experimental error, the principal strain directions (that is, the average direction of elliptical markers axes) at each site are coaxial with cleavage and lineation directions. This observation, which is consistent with those of Le Corre [1978] and Le Théoff [1977] [see Percevault and Cobbold, 1982], means that the pole to cleavage is the shortening (or minimum elongation) direction, the cleavage is the $\lambda 1\lambda 2$ principal plane, and the stretching lineation is the maximum elongation direction. These are thus the eigenvectors, or so-called principal directions, of the mean strain tensor at each site in the geographical reference frame.

The two-dimensional strain within each principal plane is classically determined by using the Rf/Φ technique of Dunnet [1969] [e.g., Dunnet and Siddans, 1971; Graham, 1978; Ramsay and Huber, 1983; Kligfield et al., 1981, 1983; Clendenen et al., 1988]. Such a technique was used here for the four sites of the preliminary study (PM, MB, LC, CB [see Cogné et al., 1986]). However, assuming that in an isotropic medium the reduction spots should have a spherical shape, it can be considered that the mean aspect ratio of elliptical spots provides an estimation of strain ratio within each principal plane [e.g., Ramsay and Huber, 1983, p. 73]. This assumption is partly verified by the fact that the orientation of long axes of ellipses is roughly constant within each studied surface. Consequently, a simple arithmetic mean of the measurements of axial ratios within each section (typically 20 to 30 measurements per section) was used in the remaining sites to estimate the axial ratio of two-dimensional strain. Standard deviations of the mean (σ) are used as error bars on these estimates. In the sites where strain ratios were determined from Rf/ϕ data, error bars have been roughly estimated from the range of Rf/Φ curves that can be superimposed on data.

Principal strains $\lambda 1$, $\lambda 2$, and $\lambda 3$ have been calculated from the two ratios $\lambda 1/\lambda 2$ and $\lambda 2/\lambda 3$, assuming there have been no volume changes during deformation, i.e., $\lambda 1\lambda 2\lambda 3 = 1$. From the diagonal matrix S'_{ij} of principal strains, and the transformation matrix a_{ij} of principal directions [e.g., Bourne and Kendall, 1977], the symmetrical strain tensor of each site can then be computed following the transformation rule for second order tensors:

$$S_{ij} = a_{ik} \cdot a_{jm} \cdot S'_{km} \quad (i, j, k, m = 1, 2, 3)$$

Anisotropy of magnetic susceptibility was measured on standard paleomagnetic specimens (cylinders 21 mm long and 25 mm in diameter) using the Digico anisotropy delineator [see Collinson, 1983] improved by the addition of a filter/amplifier on the pick-up coils signal circuit, and calibrated following the correction for instrumental error as proposed by Hrouda et

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TABLE 1. Mean-site Structural Data in the Pont-Réan Formation

Site	S_0		S_1		L		λ_1	λ_2	λ_3	K	r
	S	D	S	D	De	I					
SA	51	35	262	81	79	18	1.400	1.000	0.714	1.00	1.80
SB	269	38	277	80	278	7	1.486	1.189	0.566	0.23	2.35
SD	280	19	57	82	59	13	1.527	0.848	0.771	8.00	1.90
SE	299	30	252	88	71	23	1.472	1.090	0.623	0.46	2.10
SF	146	13	77	90	257	12	1.191	0.916	0.916	-	1.30
SG	87	20	87	90	87	0	1.278	1.065	0.730	0.45	1.65
SH	90	26	270	71	90	0	-	-	-	-	-
SI	112	9	261	81	262	4	1.339	0.864	0.864	-	1.55
SJ	90	37	278	79	97	5	1.652	1.180	0.513	0.31	2.70
MB	66	28	88	86	89	12	1.449	1.115	0.619	0.37	2.10
PM	56	33	91	80	95	22	1.206	1.096	0.756	0.22	1.55
LC	135	22	69	86	247	20	1.392	0.928	0.774	2.50	1.70
CB	280	56	101	86	101	0	1.618	1.471	0.420	0.04	3.60

Bedding (S_0) and cleavage (S_1) planes are given as S, strike; D, dip (downward in the direction S+90°); lineation (L) is given as De, declination; I, downward inclination; λ_1, λ_2, and λ_3 are principal strains,estimated from axial ratio measurements of reduction spots; K, shape parameter, $K = (\lambda_1/\lambda_2 - 1)/(\lambda_2/\lambda_3 - 1)$; r: intensity parameter, $r = \lambda_1/\lambda_2 + \lambda_2/\lambda_3 - 1$.

al. [1983] and Veitch et al. [1983]. The k_{zz} component of the susceptibility tensor (i.e., the susceptibility along the axis of the cylindrical specimen) was measured with the Digico bulk susceptibility meter. From the N measured specimens at each site, a mean-normalized susceptibility tensor has been computed by using the tensor statistics of Jelinek [1978] derived from Hext [1963]. Following this analysis, each tensor determined by a measurement is expressed in a common reference frame (e.g., geographic) and is normalized to its principal invariant $(k_{11} + k_{22} + k_{33})/3$, in order to eliminate the effects of the fluctuating specimen bulk susceptibility upon the estimation of the site-mean tensor. The mean tensor and its confidence limits are then estimated by summing the N normalized tensors of a site and computing the relevant covariance matrix, from which are derived the variance (σ^2) of the normalized principal susceptibilities and the semi-angles (a, b) of the ellipse of confidence at the 95% probability level around the principal directions. To complete the description of AMS, the mean-site bulk susceptibility is calculated as $k = (k_1 + k_2 + k_3)/3$, where k_1, k_2, k_3 are the eigenvalues of the unnormalized mean tensor (computed as above, but without normalizing each tensor by its principal invariant $(k_{11} + k_{22} + k_{33})/3$)

Results

Mean structural data obtained in the Pont-Réan red beds are given in Table 1: bedding (S_0) and cleavage (S_1) are characterized by their strike (S, declination of the horizontal line of the plane) and dip (D, maximum downward inclination of the plane in the direction S+90°). The stretching lineation L (De, declination; I, inclination), as observed in the field and verified from elliptical marker measurements on oriented blocks, is systematically at the bedding/cleavage intersection. Strain estimates are given in Table 1 as λ_1, λ_2, λ_3, principal strains; K, r: shape [Flinn, 1962] and intensity [Watterson, 1968] parameters. Mean AMS data are given in Table 2 where each normalized principal susceptibility k_1, k_2, k_3, is characterized by its intensity and standard deviation, its declination and inclination (D/I), and the 1/2 aperture angles (a/b) of the 95% confidence ellipses around the mean directions. The bulk susceptibility is given in dimensionless SI units, %An is $(k_1/k_3-1)\times100\%$ [Van Der Voo and Klootwijk, 1972].

The principal directions of strain and AMS are shown in Figure 2. The homogeneity of AMS within each site is underlined by the generally small dimension of the 95% ellipses of confidence around principal susceptibility directions. Upon examining the data, we see that the maximum susceptibility k_1 is generally subcoaxial with the bedding/cleavage intersection, that is, with the maximum elongation axis λ_1. The minimum susceptibility k_3 is, on the average, parallel to the pole to cleavage, λ_3. There are, however, some significant differences in sites SG and SI, where k_2 and k_3 show intermediate directions between pole to cleavage and pole to bedding. This tends to indicate that magnetic fabrics average the effects of the superimposition of tectonic strain upon an initial compaction- induced fabric [Graham, 1978; Ramsay and Huber, 1983]. Such an interpretation is corroborated by the exceptionnal data of site SF, where although cleavage is already observable, k_3 is not yet reoriented toward cleavage pole but remains parallel to bedding pole. It can further be noted

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TABLE 2. Mean-site Normalized Susceptibilities in the Pont-Réan Formation

Site	N	Bulk	k1				k2				k3				%An
		x10^{-3} SI	Int.	σ	D/I(°)	a/b(°)	Int.	σ	D/I(°)	a/b(°)	Int.	σ	D/I(°)	a/b(°)	
SA	32	0.126	1.023	0.006	82.5/15.5	3/2	0.992	0.003	235.0/72.5	10/3	0.984	0.005	350.5/ 8.0	10/3	3.9
SB	19	0.149	1.022	0.005	100.0/ 4.5	15/3	1.017	0.004	24.5/72.0	15/3	0.960	0.008	189.0/17.0	4/2	6.4
SD	32	0.143	1.030	0.007	60.0/12.0	4/2	0.992	0.003	282.5/74.0	6/2	0.977	0.008	152.5/10.0	6/3	5.4
SE	23	0.125	1.023	0.003	81.0/ 3.0	11/6	1.008	0.005	338.0/76.5	11/2	0.968	0.005	171.5/13.5	6/2	5.7
SF	25	0.115	1.012	0.003	258.5/13.0	5/3	1.002	0.003	166.5/ 8.1	5/4	0.985	0.004	46.0/74.5	4/3	2.7
SG	22	0.115	1.014	0.002	89.0/ 3.5	6/2	0.997	0.001	181.5/40.0	10/4	0.988	0.003	355.0/49.0	9/2	2.6
SH	23	0.109	1.018	0.003	284.0/13.5	5/3	0.997	0.004	71.0/73.0	10/3	0.984	0.003	182.0/ 8.5	10/4	3.4
SI	15	0.113	1.020	0.003	271.0/ 1.0	4/3	0.993	0.003	180.0/49.0	20/4	0.987	0.002	1.5/41.0	20/3	3.3
SJ	20	0.076	1.029	0.005	103.0/ 2.0	4/1	0.997	0.005	3.5/79.0	5/2	0.973	0.005	193.5/10.0	5/3	5.8
MB	15	0.104	1.019	0.005	98.0/ 2.5	7/4	1.004	0.003	350.0/83.0	6/4	0.976	0.005	188.5/ 7.0	7/5	4.4
PM	13	0.146	1.026	0.003	74.5/ 6.5	7/4	0.996	0.006	244.0/83.0	12/6	0.978	0.008	344.5/ 1.0	12/5	4.9
LC	6	0.110	1.040	0.011	255.0/21.0	12/6	0.984	0.007	109.0/65.0	55/7	0.976	0.009	350.0/13.0	55/7	6.6
CB	8	0.154	1.046	0.016	99.0/19.0	19/3	1.018	0.012	273.5/71.0	19/10	0.936	0.020	8.5/ 2.0	13/6	11.8

Principal susceptibility values (Int.) are given with their standard deviation (σ), and 1/2 aperture angles (a/b) of 95% confidence ellipses around principal directions. D/I, declination/ inclination. %An is $(k_1/k_3-1)\times100$%. "Bulk" is computed as $(k_1+k_2+k_3)/3$, where k_i are the principal susceptibilities of unnormalized mean tensors.

that in this site, as well as in site LC, where ellipses around k_2 and k_3 show that these directions are ill-defined in the k_2k_3 plane, the structure is of pencil type [Graham, 1978], that is, the fabric is macroscopically constrictive.

The ellipsoid shape and intensity of mean strain and AMS are illustrated by the Flinn [1965] diagrams of Figure 3. Points are distributed in the constriction and flattening areas for strain as well as for AMS. Intensities are quite low with $r < 2.5$ and %An <6% in most sites. Greater intensities are reached for points situated in the flattening field (sites SB, CB for example). In Figure 3, a clear difference appears between the shape of strain ellipsoids and the shape of AMS ellipsoids. As a matter of fact, AMS ellipsoids are, on the average, more constrictive than strain ones (see for example, sites PM, SF, SG). Two main reasons can be evoked to explain this difference. The first one is that magnetic fabric recorded the compaction of red beds stronger than reduction spots did. It could be inferred that reduction spots develop during late stages of compaction. A second and equivalent reason could be that AMS evolution is slower than total strain evolution as estimated from reduction spots. In both cases, strain ellipsoids should be more flattened than AMS ones, during progressive deformation, following the path described by Graham [1978] or Ramsay and Huber [1983]. It is, however, difficult to stress the actual physical reason for this difference, and even to assess its significance. As a matter of fact, if we take into account the low intensities of strain and AMS, small variations of intermediate-axis values (λ_2 or normalized k_2) around unity can induce significant variations of relative values between λ_1/λ_2 and λ_2/λ_3, and between k_1/k_2 and k_2/k_3. Therefore if we attempt to draw a correlation between strain and AMS ellipsoids shapes and intensities, it appears better to look for a correlation between principal values, rather than between principal ratios.

To this purpose, and following the relationship proposed by Kligfield et al. [1981, 1983], we have computed the linear regression between the principal susceptibility differences, $M_i=(k_i-k_o)/k_o$ with $k_o = (k_1.k_2.k_3)^{1/3}$, and natural strains, $\varepsilon_i=\ln(\lambda_i)$. This is shown in Figure 4. The correlation appears quite good (correlation coefficient $r = 0.943$), and the best fit line $\varepsilon_i=13.6\ M_i-0.04$ goes through the origin (Figure 4). This means that variations of AMS are, on the average, controlled by strain variations. The remarks made above, about the distribution of data in Flinn diagrams (Figure 3), appear justified, and are verified by the distribution of intermediate values (λ_2 and M_2: triangles) around zero in the diagram of Figure 4.

Finally, all the above observations can be classically interpreted in terms of progressive deformation, as beautifully described by Graham [1978] in the Maritime Alps red beds. In the initial sedimentary state, compaction induces a bedding fabric showing oblate ellipsoids with λ_3 coaxial with the pole to bedding. This state is not actually observed in the Pont-Réan Formation but can be inferred from the orthogonality of k_3 and bedding in site SF. During ductile tectonic

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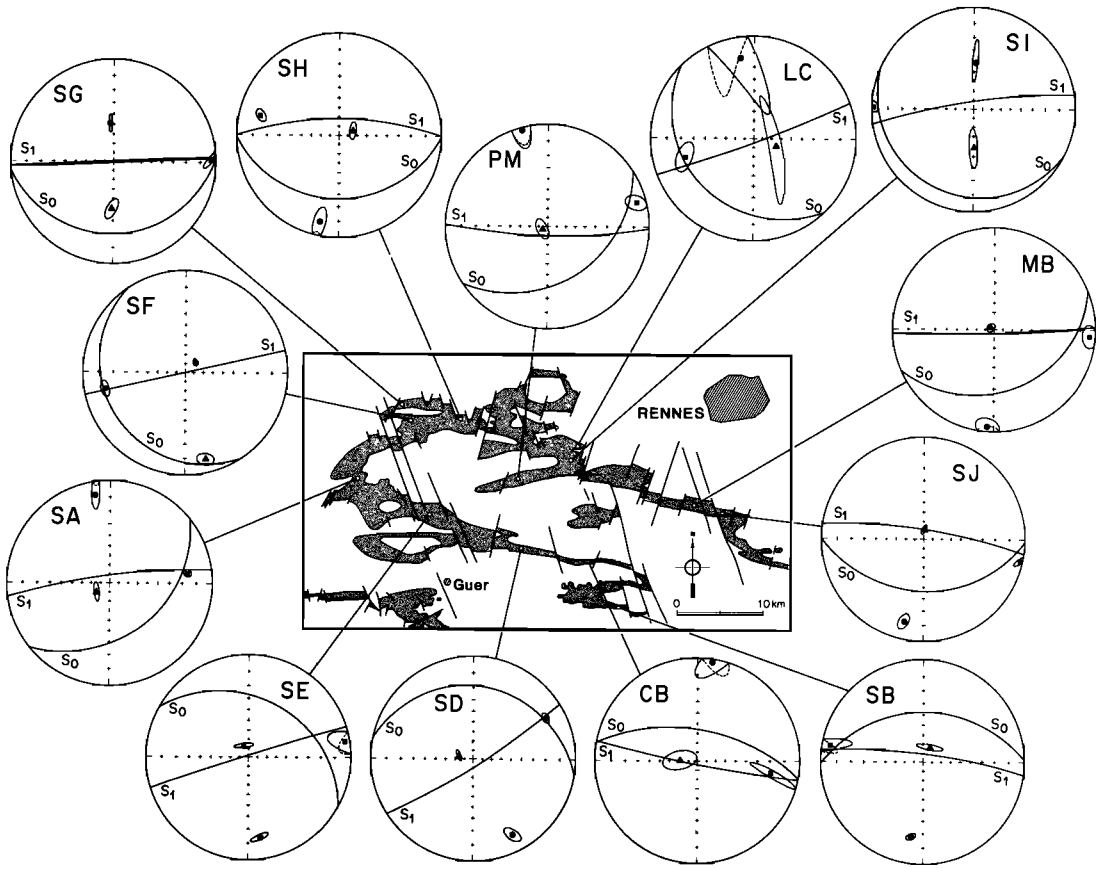


Fig. 2. Stereographic projections in the lower hemisphere of mean-site structural and AMS directions. S_0, bedding; S_1, cleavage; stretching lineation is at the bedding/cleavage intersection. Principal susceptibilities with their 95% confidence regions are squares, k_1 (max); triangles, k_2 (intermediate); and circles, k_3 (min).

deformation, the regionally subhorizontal shortening results in a constrictive shape of the total strain ellipsoids with joint development of stretching and maximum susceptibility at the bedding/cleavage intersection. Then, with increasing deformation, ellipsoids return in the flattening field with λ_3 as the normal to cleavage. This sequence is accompanied by a progressive resetting of the minimum susceptibility k_3 from the normal to bedding (site SF) to the normal to cleavage (sites SA, SB, SD, SE, SH, SJ, PM, MB, CB), through intermediate magnetic fabrics (sites LC, SG, SI).

One can further note that this sequence roughly describes a strain gradient from north of the studied area toward the south. This results in lower strain intensities in the nothern part (see Table 1; sites SF, SG, SI) than in the southern part (sites CB, SB, SJ, MB). This observation is consistent with the description of Gapais and Le Corre [1980] and Percevault and Cobbold [1982], and with their interpretation that most of Hercynian folding of Paleozoic formations in Central Brittany is controlled by dextral transcurrent movement on crustal faults.

Paleomagnetic Analysis

Natural Remanent Magnetization

Natural remanent magnetization (NRM) intensities (Figure 5) range between 0.001 and 0.01 A/m, which are typical values for red beds. Although weak, these intensities are sufficient for the remanent magnetization to be analyzed through demagnetization procedures, and measured using the Schonstedt DSM-1 spinner magnetometer (sensitivity 10^{-9} Am^2). NRM directions, shown as in situ in the density plot of Figure 5, display an elongated path between SSW slightly inclined directions toward WSW highly inclined ones. With reference to paleomagnetic data from the Armorican Massif [Perroud, 1985], it can be surmised that this distribution arises from a partial remagnetization of primary highly inclined Ordovician directions, in the SW slightly inclined Carboniferous magnetic field, during the Hercynian orogenesis.

Demagnetization

In order to test for the origin and nature of NRM, vector analysis of magnetization through stepwise thermal demagnetization was conducted on 100 specimens. The behavior of magnetization during demagnetization was generally simple. Two types of demagnetization curves (in orthogonal projections [Zijderveld, 1967]) were obtained (Figure 6). (1) On the one hand, when NRM initially exhibits high inclinations, the magnetization intensity progressively decreases with increasing temperature, without any change in direction (Figure 6a). This is characteristic of univectorial magnetization. One can note that

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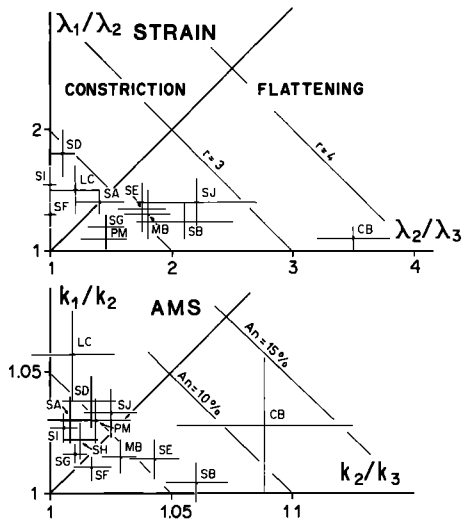


Fig. 3. Flinn [1965] diagrams for mean strain and AMS data. Lines of strain intensities r=2, 3, and 4, and of %An = 5, 10, and 15% are drawn. Error bars on strain ratios are the standard deviations on arithmetic mean; error bars on susceptibility ratios are derived from standard deviations on mean susceptibilities [after Jelinek, 1978].

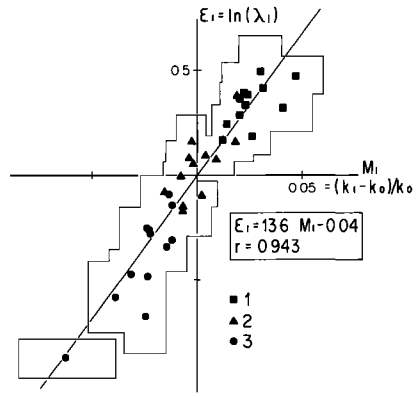


Fig. 4. Principal susceptibility differences (M_i) versus natural strains (ε_i). The best fit line is shown and regression equation over points is given. Error grid is derived from standard deviations on mean susceptibilities and strain axial ratios. Same symbols as in Figure 2.

unblocking temperatures are separated in two groups: a part of the magnetization is progressively unblocked between 20° and 600°C, then it stabilizes between 600° and 650°C, and is finally unblocked between 650° and 680°C, the Curie point of hematite. (2) The second kind of behavior was observed mainly on specimens where NRM was initially either slightly or intermediately inclined SW. It is characterized by two magnetization components (Figure 6b): a low-temperature (LT) component which is unblocked up to about 600°C and shows a shallow southwesterly direction. The second magnetization component is more steeply inclined W to SW, and appears similar to the magnetization component described in the case of univectorial behavior. This high-temperature (HT) component is unblocked in the 650-680° temperature range. Note, however, that a good separation of these components is not always ensured, as shown in Figure 6b, specimen SA04. A separation into such LT and HT components was obtained in five sites (SA, SD, SH, SI, LC); in other sites, a unique steeply inclined magnetization, covering the whole temperature spectrum, is dominant.

The bimodal distribution of unblocking temperatures points to the presence of two main magnetic carriers. However, some isothermal remanent magnetization (IRM) experiments allowed identification of only one kind of magnetic mineral, with high coercivities, typical of hematite. The range of unblocking temperatures of LT components is consistent with ultrafine red hematite pigment, while high temperatures could be related to larger hematite crystals, possibly specularite. Following this interpretation, shallow LT components could result from chemical remobilization of pigment, more sensitive to fluid circulation during the Hercynian orogeny, and could thus actually be secondary.

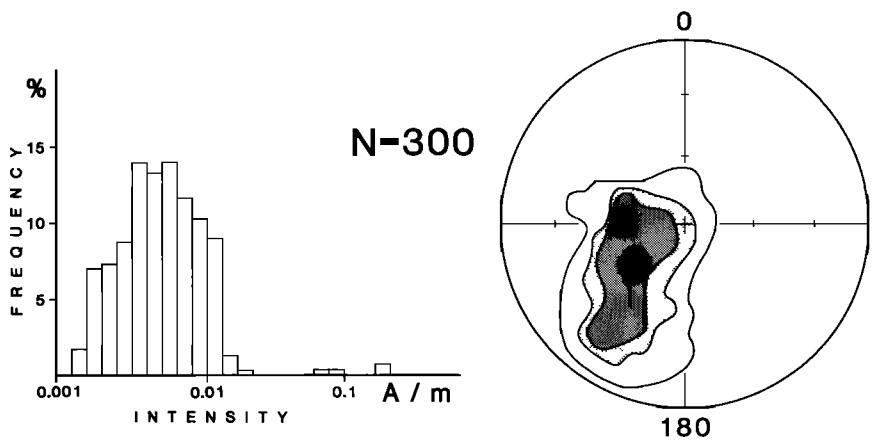


Fig. 5. (Left) Frequency plot of NRM intensity. (Right) Equal-area density plot in the lower hemisphere of NRM directions in geographic coordinates; contours: 1, 2.5, 5, and 10% of the data in 1% of the hemisphere surface.

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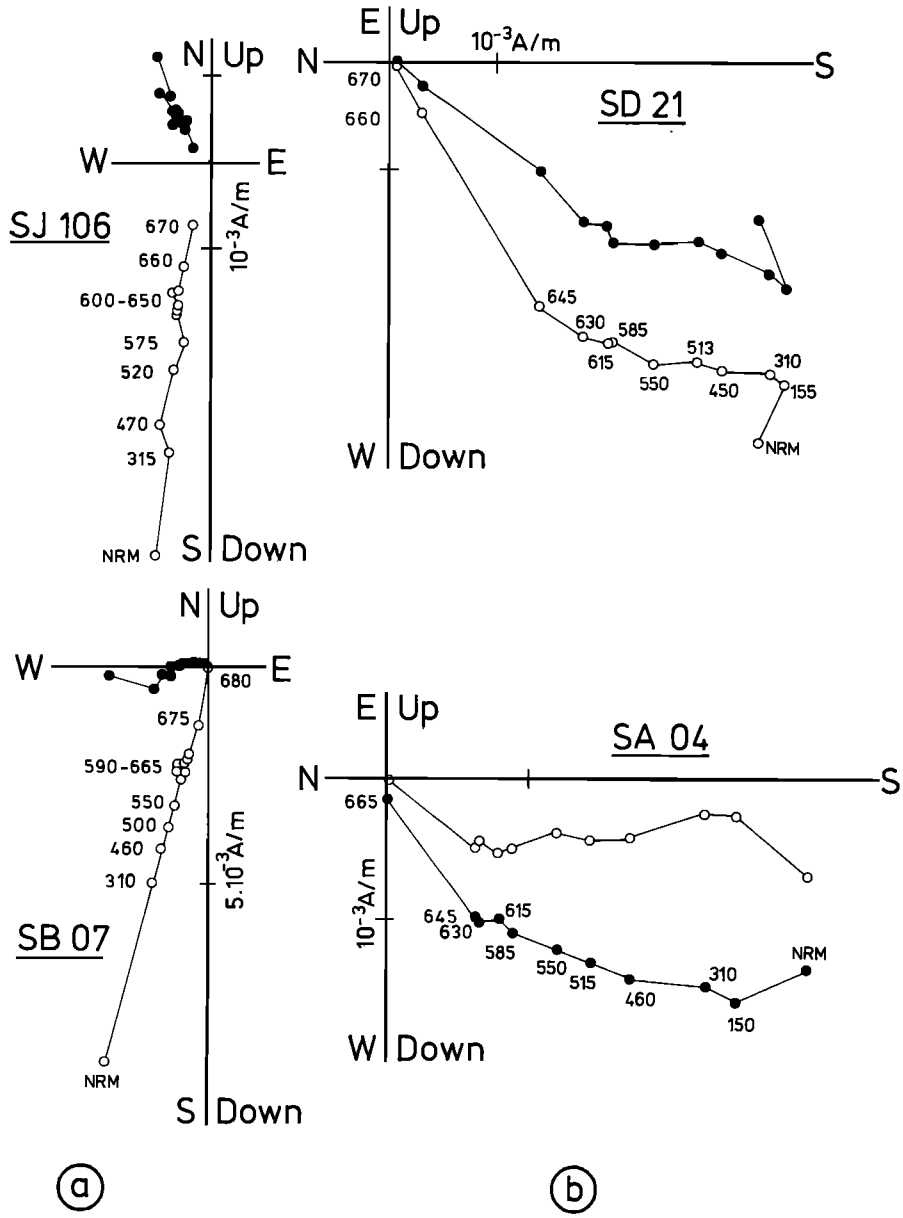


Fig. 6. Thermal demagnetization of representative specimens in Zijderveld [1967] orthogonal projections. Closed (open) symbols are projection onto the horizontal (vertical) plane. For explanation of Figures 4a and 4b, see text.

Analysis of Remanent Magnetization Directions

Low-temperature components. Mean-site directions of LT components are listed as in situ (IS) and tilt corrected (TC) in Table 3 and drawn in Figure 7. In the in situ coordinates, these directions show a quite constant declination of about 200°, but a great scatter in inclination. This dispersion is not reduced by the classical tilt correction [Graham, 1949]. This situation can partly arise from a synfolding age of magnetization acquisition. However, in the in situ coordinates, points show a great-circle distribution between the steeply inclined formation mean of high-temperature components (Table 4) and shallow Carboniferous and Lower Permian dipole field directions (Figure 7). LT magnetization components are thus probably syntectonic to post tectonic partial remagnetizations, but poorly separated by thermal demagnetizations, due to the overlapping of blocking temperature spectra. This bad separation does not allow us to compute an accurate mean direction of magnetization for this secondary component.

High temperature components. Mean-site direction of magnetization HT components are quoted in Table 4 and drawn in Figure 8 as in situ (IS) and classically tilt corrected (TC). As noted above, these data display W to SW declinations and steep downward inclinations. Such directions are quite similar to the

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TABLE 3. Mean-site Magnetization Low-Temperature Components Isolated by Thermal Demagnetization

Site	N	IS		TC		k	α_{95}
		De	I	De	I		
SA	7	201	3	204	-14	43	9
SD	5	200	6	201	25	49	11
SH	7	202	23	200	-2	58	8
SI	5	183	54	186	45	41	12
LC	6	206	31	208	10	44	10

IS, in situ; TC, tilt corrected; De, declination; I, inclination; k and α_{95}, Fisher's [1953] statistics parameters; and N, number of entries in the statistics.

characteristic remanent magnetization direction (RM) obtained by Perroud et al. [1986] in the Ordovician "Moulin de Chateaupanne" red formation (D = 228°, I = 81°), and by Perroud and Van der Voo [1985] in the the Late Ordovician intrusive Thouars Massif (reversed direction D = 160°, I = 83°). In the IS projection of Figure 8, our directions are also clearly different from the present-day dipole field axis. Since we do not know any steeply inclined field direction in the Armorican Massif since Ordovician times [Perroud, 1985], it is assumed that these high unblocking temperature components, isolated by thermal demagnetization, are actually the primary pretectonic magnetization component of the Pont-Réan Formation.

Upon examining the data of Table 4 and Figure 8, two remarks can be made.

1. At the scale of each site, magnetization vectors are extremely well clustered, with the exception of sites CB and SA, which will be discussed later. The k parameter of the Fisher's [1953] statistics of the within-site distributions is often higher than 100 (sometimes 200), and in the case of site SD, it is higher than 1000. Such values, which are quite common within volcanic rocks where thermoremanent magnetization is an instantaneous recording of magnetic field direction, can be regarded as anomalously high for a sedimentary formation where within-site k values classically range from 50 to 100.

2. The second remark deals with the comparison of in situ and tilt- corrected between-site distributions (Figure 8). It is obvious that the simple tilt correction induces a large scattering of the quite well-clustered in situ data, in relation with the structural position of sites in the southern or northern limb of the syncline. Following the classical interpretation of the fold test, this should lead us to interpret these paleomagnetic directions as postfolding. We have thus an anomalous behavior of magnetization during unfolding, and the same paradoxical situation as in the Maritime Alps: tilt correction tends to disperse data which are not interpretable as synfolding or postfolding remagnetizations.

Strain-induced deviation of RM. Here, as in previous cases, these anomalies can be suspected to arise from a deviating effect of strain upon the high unblocking temperature component of remanent magnetization (RM). The proposed interpretation is shown schematically in Figure 9. The sedimentary beds are supposed to be initially subhorizontal and have a subvertical RM. Strain during folding of the formation induces the development of a subvertical axial planar cleavage. Taking into account the high angle between the subhorizontal shortening direction and the subvertical initial magnetization, the deviation of remanent vectors toward the flattening plane (the cleavage) results in both effects described above.

1. At the scale of each site where strain is homogeneous (Figure 9b) the deviation of magnetization is achieved without any loss in within-site grouping of magnetization, as was the case in the Maritim Alps red beds [Kligfield et al., 1981, 1983; Cogné and Perroud, 1985]. In contrast, deformation probably tends to cluster the data, as was observed in experimental deformation of synthetic samples [Cogné, 1987b], when such a case of a high angle between shortening direction and initial direction of magnetization was simulated. The anomalously high clustering of within-site distributions described above can thus be explained as the result of strain acting on pretectonic remanent magnetization.

2. Since the remanent magnetization is initially close to the flattening plane, the deviation towards cleavage induces an overall cluster of magnetization vectors at the scale of the folded area (Figure 9b). This explains the good grouping of data in the in situ coordinates, and the inevitable dispersion when applying the classical tilt correction (Figure 9c).

Obviously, this highly schematic description is convenient only to explain main features of the results that are (1) the high within-site cluster of paleomagnetic data and (2) the good grouping of between-site distribution in the in situ coordinates. However, the simplifying assumptions used in the description of Figure 9

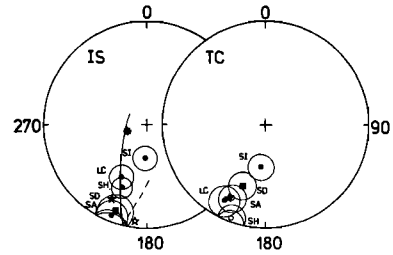


Fig. 7. Mean-site direction of low-temperature components with their circles of 95% confidence. IS, in situ; TC, tilt corrected. In the IS projection, asterisk and closed and open stars are the formation mean of HT components (quoted in Table 4) and the Carboniferous and the Lower Permian field directions, respectively; the great circle is the best fit plane over these three directions. Closed (open) symbols are projection in the lower (upper) hemisphere. Stereographic projections.

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TABLE 4. Mean-Site High-Temperature Components of Magnetization Isolated by Thermal Demagnetization in Geographic (In Situ) and Bedding (Unfolded) Coordinates, and After Strain Removal and Tilt Correction from the Unstrained Bedding Plane (Undeformed)

Site	n/N	In Situ		Unfolded		Undeformed		k	α_{95}
		De	I	De	I	De	I		
SA*	6/8	248	40	216	42	-	-	34	10
SB	7/8	272	82	347	51	268	75	268	4
SD	7/8	218	46	234	62	202	70	1008	2
SE	8/8	257	56	310	64	231	85	178	4
SF	7/8	229	79	233	66	233	74	211	4
SG	8/8	263	71	220	62	227	69	115	5
SH	7/8	254	73	209	55	188	56	103	6
SI	8/8	264	65	249	60	251	67	100	5
SJ	7/9	289	70	214	55	184	60	79	7
MB	9/9	264	58	220	56	205	57	79	6
PM	6/6	284	55	223	66	238	81	213	4
LC	6/6	223	61	224	40	202	54	168	5
CB*	5/6	217	63	348	57	-	-	23	16
Mean	11/13	254	67	237	64	213	70		
k		29		13		35			
α_{95}		8		13		8			

n/N, ratio of number of entries in the statistics to number of analyzed specimens; De, I, k, and α_{95}, same as in Table 3.
*Data rejected from the formation mean estimations (mean).

(initially horizontal bedding plane and vertical magnetization, horizontal shortening) are far from being exactly obeyed in nature. One discrepancy between the predicted and real final distributions must be particularly noted. From the schematic description of Figure 9, we should obtain the following features (Figure 10a): in the south dipping beds of sites from the northern limb, paleomagnetic vectors should be located on the northern side of the cleavage (or magnetic foliation) plane, while they should be in the southern side of this plane in the sites from the southern limb. Although this situation is generally met in the latter case (sites SE, SD, CB, SB), this is not not the case for some sites from the northern limb (Figure 10b). This is particularly obvious in sites LC, MB, SH, and SJ. Therefore these four sites do not obey the global behavior described above. However, this anomaly disappears if we consider that magnetization was not acquired by horizontal beds but by already southerly dipping beds. This hypothesis is

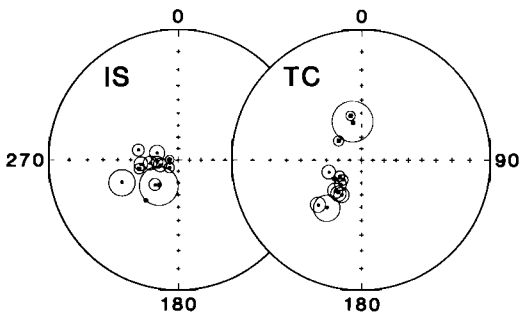


Fig. 8. Mean-site directions of high-temperature components with their circles of 95% confidence. Squares show directions from the southern limb sites. Same conventions as in Figure 7.

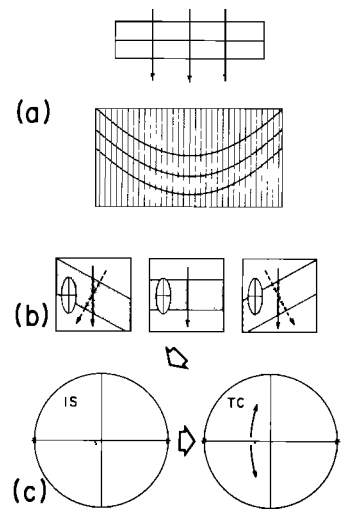


Fig. 9. Schematic interpretation of strain effects upon primary magnetization within red beds of the Pont-Réan Formation. For explanation of Figures 9a, 9b, and 9c, see text.

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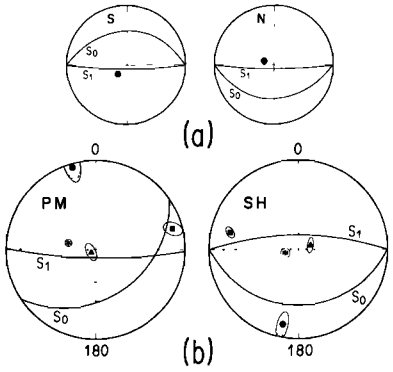


Fig. 10. (a) Theoretical angular relationships of mean-site paleomagnetic vector (star), bedding (S_0), and cleavage (S_1) for sites from the northern (N) and southern (S) limbs of the syncline. (b) Observed relationships in two sites (PM and SH) from the northern limb. AMS directions with their ellipses of 95% confidence are drawn with same conventions as in Figure 2.

consistent with the description by Ballard et al. [1986] (see Figure 1d) of a synsedimentary dip of bedding planes toward the center of basin, and will be further verified by the results of strain removal applied to these paleomagnetic vectors, as discussed at the end of this paper.

With respect to the global behavior described above, the weak within-site grouping of sites SA and SB can be considered as anomalous. However, HT magnetization components of both sites (Figure 11) show a non-Fisherian, elongated path from high inclinations toward SW lower inclinations. It is thus probable that primary Ordovician and secondary Carboniferous magnetization components have been poorly separated by thermal demagnetization within these specimens, as shown in Figure 6b. Since they are probably badly separated directions, the poor quality of their within-site cluster does not interfere with the above analysis, and the data from sites SA and CB were rejected from the following.

Mechanism of RM deviations. The in situ distribution of paleomagnetic vectors is here in sharp contrast to what was observed in the Maritime Alps red beds, where strain has a scattering effect upon populations of remanent magnetization vectors. We must, however, point out that it is the same fundamental mechanism that causes such different final states: the deviation of specimen magnetization toward the strain flattening plane. As described earlier [Cogné and Perroud, 1987], the deviation of within-site directions and their change in grouping then depend upon the angle between strain shortening axis and initial magnetization direction at the scale of each site of homogenous deformation, while the overall distribution of mean-site directions depends upon strain distribution across the folded area. These observations point to the need of considering first the effects of deformation at a scale where strain is homogeneous, before attempting to describe them at the scale of a whole folded formation, where strain is typically heterogeneous. The only common point at the latter scale is the failure of the classical fold test when using paleomagnetic data from strained folded beds.

The rotation of net measurable remanent magnetization probably results from the rotation of elementary magnetic micromoments and thus from the rotation of hematite particles that carry them. This assumed mechanism is in good agreement with the analysis of AMS data made above. As a matter of fact, if we consider that AMS is mainly dominated by hematite anisotropy, the good correlation established between AMS and strain shows that strain controls the development of hematite basal plane preferred orientation. This preferred orientation development may be achieved in part through recrystallization of oriented hematite, as underlined by the occurrence in some sites of a secondary magnetic overprinting related to the Hercynian orogeny. On the other hand, there is no clear difference in strain/AMS relationships in sites showing only the primary pretectonic remanent magnetization. It seems therefore reasonable to assume a rotation mechanism for the reorientation of primary hematite particles which carry the pretectonic remanent magnetization. Note, however, that with the evidence at hand, it is not possible to state categorically this rotation mechanism for primary hematite. Nevertheless, the consistency of the described paleomagnetic vector behavior with previous results [Cogné and Perroud, 1985; Cogné, 1987a] where such a mechanism has been shown [Cogné and Gapais, 1986] or modeled [Cogné, 1987b] makes it highly probable.

Corrections for Strain-Induced Deviations of Pretectonic Remanent Magnetization.

Technique of Strain Removal

The analysis of pretectonic magnetization directional distribution has shown that it arises from one elementary mechanism: the deviation of specimen remanent magnetization vector from its initial direction toward the flattening plane of homogeneous internal deformation. This behavior is roughly consistent with the behavior of passive line, following the model of March [1932]. This model is thus used as a working hypothesis in order to estimate the primary direction of magnetization at the undeformed state. This technique has been proposed by Cogné et al. [1982] and described by Cogné and Perroud [1985]. It can be summarized as follows. In the theory of March [1932], the final orientation of a passive line is connected to its initial

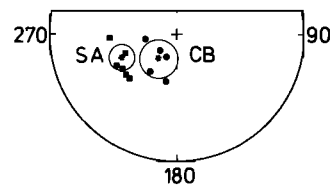


Fig. 11. Specimen HT components of magnetization and their means (asterisks) in sites SA and CB. Stereographic projections in geographic coordinates.

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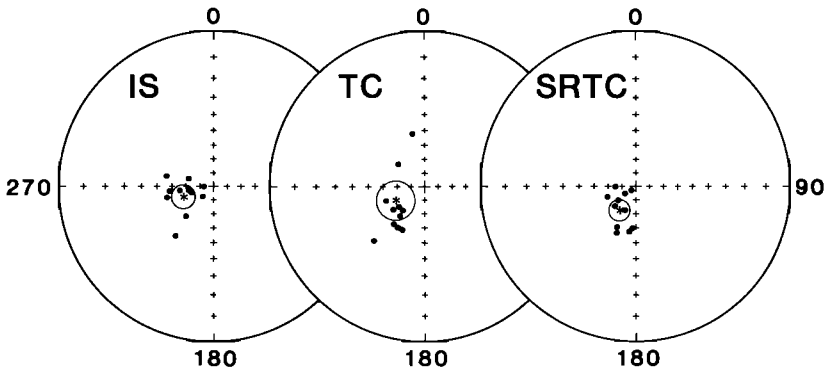


Fig. 12. Mean-site HT components of magnetization and the relevant formation means (asterisks). IS, in situ; TC, tilt corrected; SRTC, after strain removal and tilt correction from the unstrained bedding plane. Stereographic projections.

orientation through the total deformation tensor T. This tensor is decomposed into two parts [see Ramsay, 1967]: an internal deformation, or strain, tensor S, and a rigid-body rotation tensor R. In an identical way, we can decompose the total deviation of paleomagnetic vectors into two parts: (1) deviation due to strain and (2) rigid-body rotation. Following the idea that paleomagnetic vectors closely follow the model of March [1932], we assume that an estimate of the tensor of strain-induced deviation of paleomagnetic vectors is given by the strain tensor S itself. An estimation of the rigid-body rotation is provided, as in all other paleomagnetic works, by the strike and dip of bedding surface at each site. However, in deformed beds the bedding plane has also suffered from distortional strain. It is thus necessary to correct bedding tilt and direction for these effects [Owens, 1973; Cogné and Perroud, 1985]. Then, the rigid rotation is estimated from unstrained bedding planes. Note that performing the tilt correction from strike and dip of the unstrained bedding plane only corrects for the part of rigid-body rotation about the horizontal axis at each site. This method does not allow to estimate for rotations around vertical axes. Nevertheless in the present case, magnetization vectors are steeply inclined, and therefore this limitation has little effect.

To summarize, the total inverse deformation of each magnetization direction is performed in two steps: (1) unstraining paleomagnetic vector using the relevant inverse strain tensor; (2) inverse rigid-body rotation by tilt correction of the unstrained vector, using strike and dip of the unstrained bedding plane of the site.

Estimate of Mean Strain at Each Site

Measurements of AMS were used to improve the estimates of mean strain within each site by considering that mean susceptibility axes are the principal strain directions, and by resetting principal strain values from the correlation $\varepsilon_i = 13.6\, M_i - 0.04$, established above. The reason for this reestimation of strain is twofold.

1. We can consider that, on the average, principal susceptibilities and strains are coaxial over all the studied area (Figure 2). Angular departures from this coaxiality within each site are assumed to arise mainly from errors in field measurements of cleavage, lineation, etc. In contrast, the precision of core orientation using paleomagnetic sampling techniques, and the statistical processing of AMS data, suggest that principal susceptibility directions are defined with better accuracy. Therefore the eigenvectors of the strain tensor at each site have been set parallel to the eigenvectors of the AMS tensor.

2. Principal strain values have been reestimated following similar arguments. On the one hand, principal strains have been estimated from the mean axial ratio of elliptical reduction spots measured within oriented hand samples. The exact statistical meaning of such measurements with respect to the mean strain of a site is quite difficult to assess. On the other hand, paleomagnetic sampling is more widely distributed within a site, and thus it is considered to be better representative of mean AMS at each site. Here too, there is a good correlation between principal strain and susceptibility values at the scale of the whole area (Figure 4), and local departures are assumed to arise from the lower accuracy of strain determination. Principal strains were thus recalculated using the equation of regression line, and mean AMS data of each site (Table 2).

Results of Strain Removal

Mean-site directions of pretectonic remanent magnetization obtained by strain removal and tilt correction from the unstrained bedding plane are given in Table 4 and drawn in Figure 12. In this figure, they are compared with the in situ and classically tilt-corrected data. One can see that the population of corrected directions is reasonnably well clustered at the undeformed state. The paradoxical behavior of this population upon simple unfolding thus vanishes, and the hypothesis of a pretectonic magnetization is validated. The consistency of overall mean direction of magnetization at the undeformed state with Ordovician paleomagnetic direction known in the Armorican Massif [Perroud, 1985] allows us to defend the usefulness of the strain removal technique applied to the deviated

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TABLE 5. Final Estimations of Formation Mean Magnetization

	Dm	Im	k	α_{95}	N
In situ	254	67	23	13	7
Tilt corrected	258	70	13	17	7
Undeformed	235	75	87	6.5	7

remanent magnetization directions of the deformed red beds of the Pont-Réan Formation. Using the statistics of McElhinny [1964], the k parameter of the undeformed mean appears greater at the 95% probability level than in the unfolded mean, but is not different from the in situ mean. The identical grouping of in situ and undeformed means is not surprising, since the effect of strain, as described above and in Figure 9, is to cluster magnetization vectors. It is thus probable that paleomagnetic vectors have similar or better grouping after deformation than in the initial, predeformed, state.

Moreover, a more thorough examination of the results shows that unstrained paleomagnetic directions (Figure 12, SRTC) cluster into two groups. The first group of seven sites (PM, SB, SD, SE, SF, SG, SI) has inclinations higher than 70°, while in the second (LC, MB, SH, SJ), inclinations are lower than 60°. With reference to known Ordovician data, the latter appear too low, and this gives a slightly elongated distribution of points in the N-S direction. The anomaly of these four sites is related to the anomaly of their in situ data, but is easier to interpret at the undeformed state, as resulting from a bad estimation of the rigid-body rotation determined from the dip of unstrained bedding plane. In effect, as noted in the introduction, Ballard et al. [1986] have shown that the Pont-Réan formation was deposited during tilting of Brioverian blocks (see Figure 1d). Near the normal faults, the sedimentation plane may have had a significant initial dip of 10° to 20° toward the center of the basins. Sites LC, MB, SH, and SJ are actually located near such synsedimentary faults (see Figure 1b), their shift from the other group averages about 15°, and the direction of this shift toward the south is consistent with the direction of suspected synsedimentary dipping. The discrepancy of these four sites could thus arise from an incorrect tilt correction for these initially dipping sites. Note that this situation is not directly connected to our strain removal technique but to the commonly assumed, albeit difficult to verify, idea that sedimentary beds are deposited horizontally. A similar problem was encountered in the lateral equivalent "Moulin de Chateaupanne" red bed formation, where Perroud et al. [1986] showed that the Ordovician remanent magnetization was acquired by beds already tilted, probably during sedimentation.

Finally, although we have some arguments to explain the slightly deviated paleomagnetic directions of sites LC, MB, SH, and SJ, we have no precise elements to correct for these anomalies. Consequently, these four sites have been rejected from the final estimations of formation mean magnetization that are quoted in Table 5 and drawn in Figure 13.

It is the undeformed direction of Table 5 that I consider to be the characteristic pretectonic paleomagnetic direction of the Ordovician red beds of the Pont-Réan Formation. Statistical tests of McElhinny [1964] show that the k parameter of the undeformed mean is significantly higher than k of both in situ and tilt-corrected means. The undeformed direction is identical to the Ordovician magnetic dipole field direction (star in Figure 13) computed for the location 48°N, 2.5°W from the Ordovician VGP of the Armorican Massif after Perroud [1985].

Conclusions

The main results obtained in this study can be summarized as follows.

1. Folding of the Pont-Réan Formation is accompanied by the development of an axial planar cleavage. This indicates the occurrence of strain processes during folding. Strain induces the development of an anisotropy of the magnetic susceptibility. Strain and AMS tensors show a good correlation between their principal directions as well as principal values. We can thus consider that strain controls the magnetic fabric. Taking into account the presence of a pretectonic remanent magnetization, a strain-induced rotation mechanism is inferred for primary hematite that carries this magnetization. This mechanism is thought to be quite similar to the one described in the Maritime Alps red beds on the basis of X ray diffractometry measurements [Cogné and Gapais, 1986].

2. The progressive alignment of hematite within the cleavage plane induces a rotation of net measurable specimen remanent magnetization toward this plane. At the scale of each homogeneously deformed site, this results in a deviation of the mean magnetization toward cleavage. Due to the high angle between shortening and initial magnetization directions, the whole deviation is achieved with a clustering effect upon the initial dispersion. This is in close agreement with results of numerical and experimental simulations [Cogné et al., 1986; Cogné, 1987b]. At the scale of the folded area,

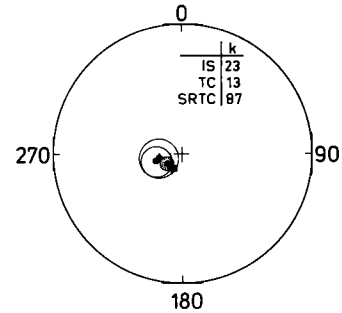


Fig. 13. Final formation mean directions for the Pont-Réan Formation. IS, TC, SRTC (dotted circle) as in Figure 11. Asterisk is the Ordovician dipole field direction in the Armorican Massif [after Perroud 1985]; k is dispersion parameter of Fisher's [1953] statistics.

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strain induces a clustering of between-site distribution of mean paleomagnetic vectors. The classical application of the fold test should have thus lead to an erroneous assignment of a syntectonic to post tectonic age to these pretectonic paleomagnetic vectors. In the terms used in the introduction of this paper, this means that the description of paleomagnetic vectors' rotation from only bedding plane geometry is, in effect, inappropriate within strained beds, and conclusions drawn from this description are inadequate. This confirms the similar analysis we made in the Maritime Alps red beds [Cogné and Perroud, 1985], and the conclusions of Van der Pluijm [1987] and Kodama [1988] drawn from numerical simulations. This point is important because we can suspect that as soon as cleavage develops, analysis of paleomagnetic vector populations with the help of the classical fold test [Graham, 1949] can be biased.

3. The probable rigid rotation of hematite particles that carry the pretectonic magnetization induces deviations of RM. These deviations appear roughly consistent with the deviations of passive lines following the model of March [1932]. The passive line model has thus been used as a working hypothesis in an attempt to correct for strain-induced deviations of paleomagnetic vectors. Application of the strain removal technique [Cogné et al., 1982] to the remanent magnetization vectors of the Pont-Réan Formation has been shown to be successful. No significant overcorrection connected with the behavior model of March [1932] has been observed. After similar results obtained in the Maritime Alps red beds [Cogné and Perroud, 1985] and the Pyrenean red beds [Cogné, 1987a], this third example confirms the usefulness of the strain removal technique in paleomagnetic studies of moderately deformed red beds.

Finally, the detailed study of strain/magnetization relationships allowed the determination of a pretectonic direction of magnetization from a deformed formation that should have been rejected by classical paleomagnetic analysis.

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